

Evaluating Climate Change Based on World Bank Data (1990-2010)

Abstract

This project examines climate change using World Bank data, focusing on six diverse countries.

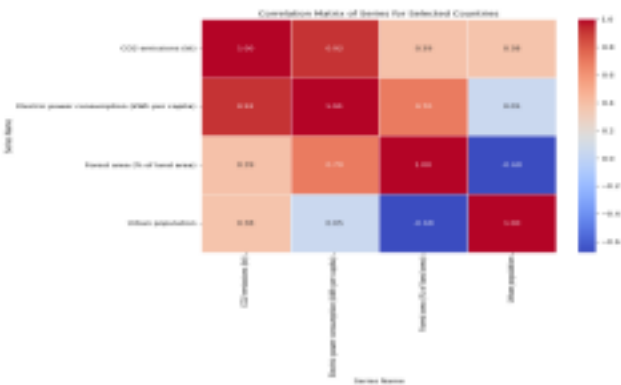
The analysis explores the relationships between CO2 emissions (kt), urban population, Electric Power Consumption (kWh per Capita, and Forest Area (% of Land Area):

Noteworthy interactions emerge, particularly in developing nations like China (CHN), Nigeria (GNA), and India (IND).

The study provides insights into the interconnected factors influencing climate change in the selected countries.

Chart. One

Correlation Matrix of Series for Selected Countries



Exploring the interconnectedness of four

nations — China (CHN), Nigeria (NGA), India (IND), and the United States (USA) — as replicated in key environmental indicators. Through the art of a heatmap, it reveals the correlations, and the hidden work that plays between these nations.

A work of numbers that calculates the relationships between the selected series. Every data entered in the matrix states how closely link or different two series are across the selected nations. The correlation values range from -1 to 1, with -1 representing a perfect negative correlation, 1 indicating a perfect positive correlation, and 0 signifying the absence of correlation.

Heatmap Unveiled: Visualizing Connections

Heatmap, a visual illustration of the correlation matrix, where the colors interpret better than words. Warm hues signify positive correlations, while cool tones hint at negative connections. Annotations

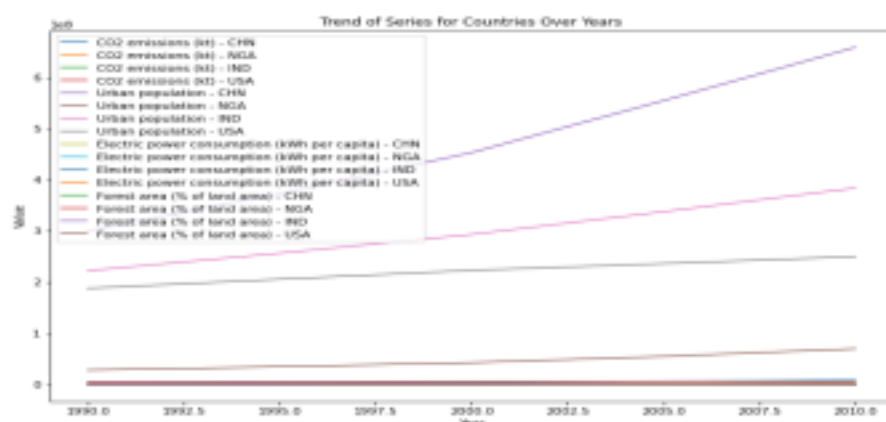
provide the numerical nuances, guiding our gaze through the intricate dance of environmental indicators.

Reflections (The Correlation):

Looking at the heatmap, correlation is seen playing across nations. A positive correlation between urban population and electric power consumption suggests the pulse of urban development. The negative correlation between forest area and CO2 emissions expresses slight tap between conservation and development.

Trend of Series for Countries Over the Years

Chart.2



The line plot tracks environmental trends (1990-2010) for China (CHN), Nigeria (NGA), India (IND), and the United States (USA) for CO2 emissions (kt), Urban population, Electric power consumption (kWh per capita), and Forest area (% of land area).

China and the USA exhibit substantial CO2 emission increases, indicating robust development, while Nigeria and India's rises are less certain.

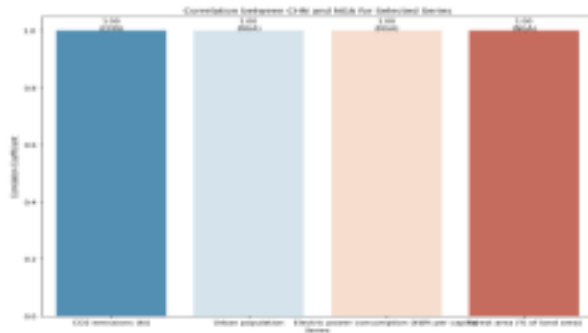
Urban population grows steadily across all countries, with significant growth in China and India, reflecting rapid urban development.

China and India show notable increases in power consumption, the USA records the highest per capita consumption, and all nations experience developmental growth.

Forest area declines in India and China due to development, Nigeria sees a similar drop, while the USA maintains a stable fraction, showcasing effective environmental management.

Chart.3

Correlation between CHINA (CHN) and NIGERIA (NGA)



The bar chart illustrates positive correlations between China (CHN) and Nigeria (NGA) across CO2 emissions (kt), urban population, Under Electric Power Consumption (kWh per Capita), and Forest Area (% of Land Area):

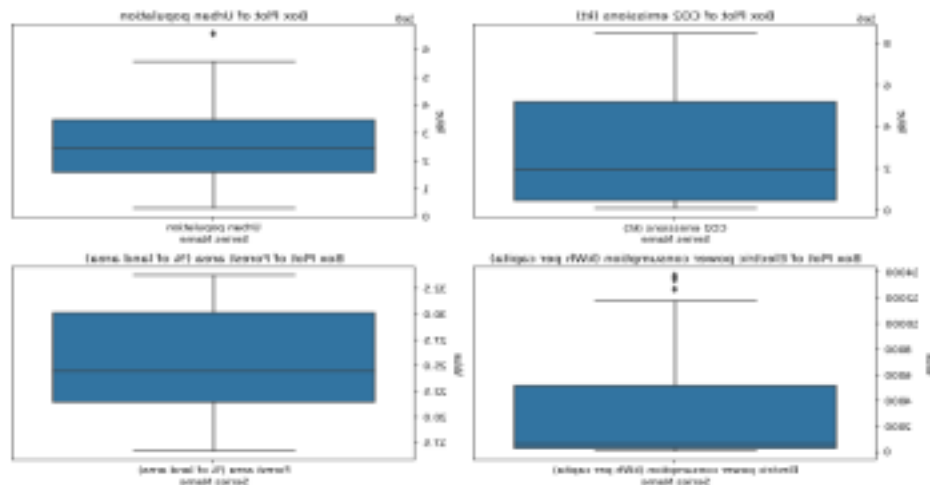
Notably, the sky-blue bars reveal synchronized increases in CO2 emissions and urban population, reflecting shared developmental trends.

Positive correlations in power consumption and forest area changes further highlight the collective environmental phase between the two nations.

The visual interpretation suggests a mutual drive and consistency in facing global ecological challenges.

Chart.4

Box Plot for different series



The 'CO2 emissions (kt)' Box Plot identifies China (CHN) and the United States (USA) as outliers, signifying significant emission differences, while Nigeria (NGA) and India show smaller boxes, indicating noticeable but less overt environmental impact.

In the 'Urban population' Box Plot, China and India display upward trends, reflecting substantial development, while Nigeria and the United States exhibit developmental growth.

'Electric power consumption' Box Plot emphasizes strong economies in China and the United States, with India and Nigeria showing less development in power consumption, reflecting preferences for electricity over time.

The 'Forest area (% of land area)' Box Plot reveals a decline in forested land for China and India due to rapid growth, Nigeria suggesting a subtle reduction, and the United States maintaining a steady portion, emphasizing preservation commitment.